BossFight

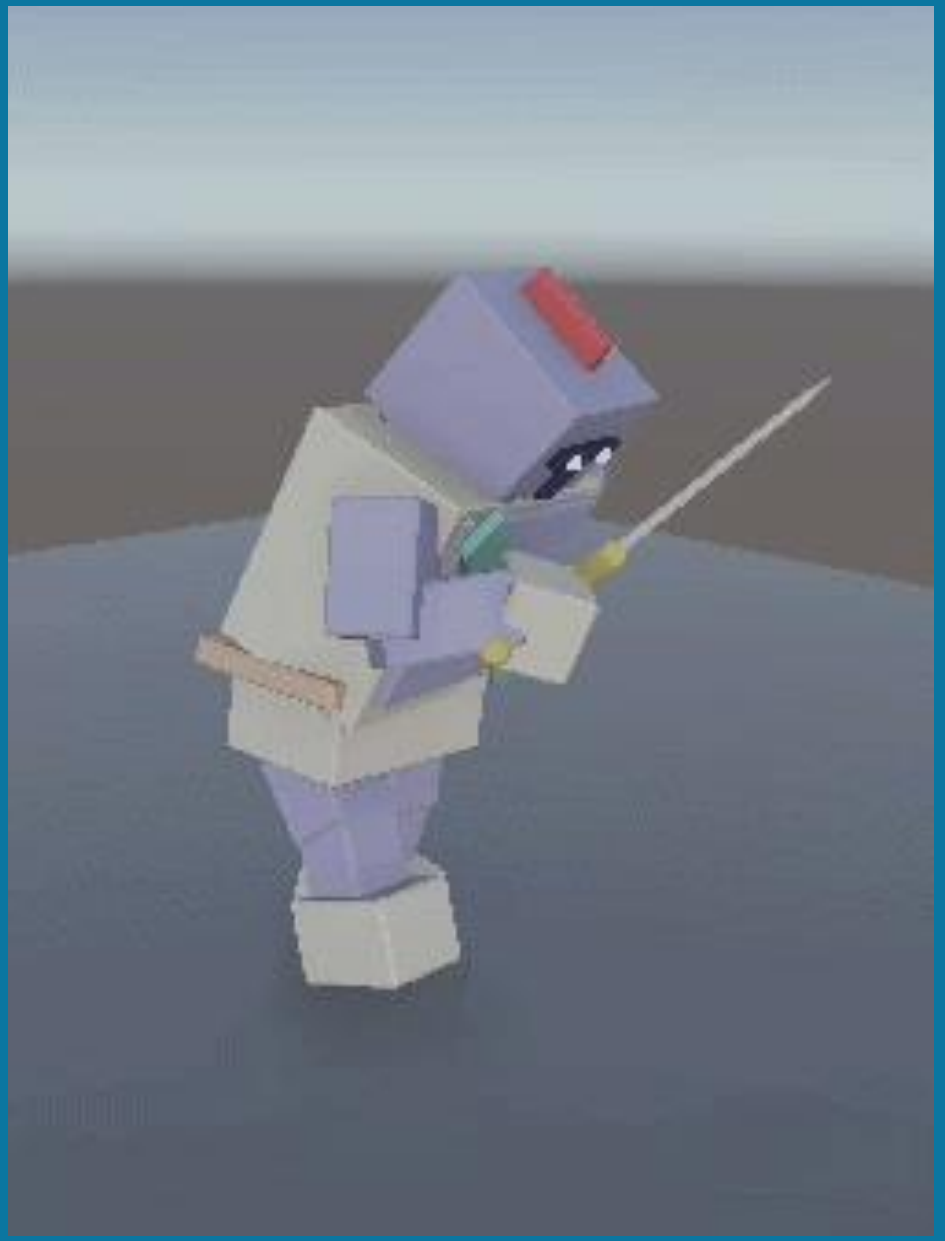
Diwen Chen

Andre Mata Assis

Richard Newman

James Prendergast

Max Wainwright



https://github.com/Maxw-Dev/MLClassProject

Proposal Recap

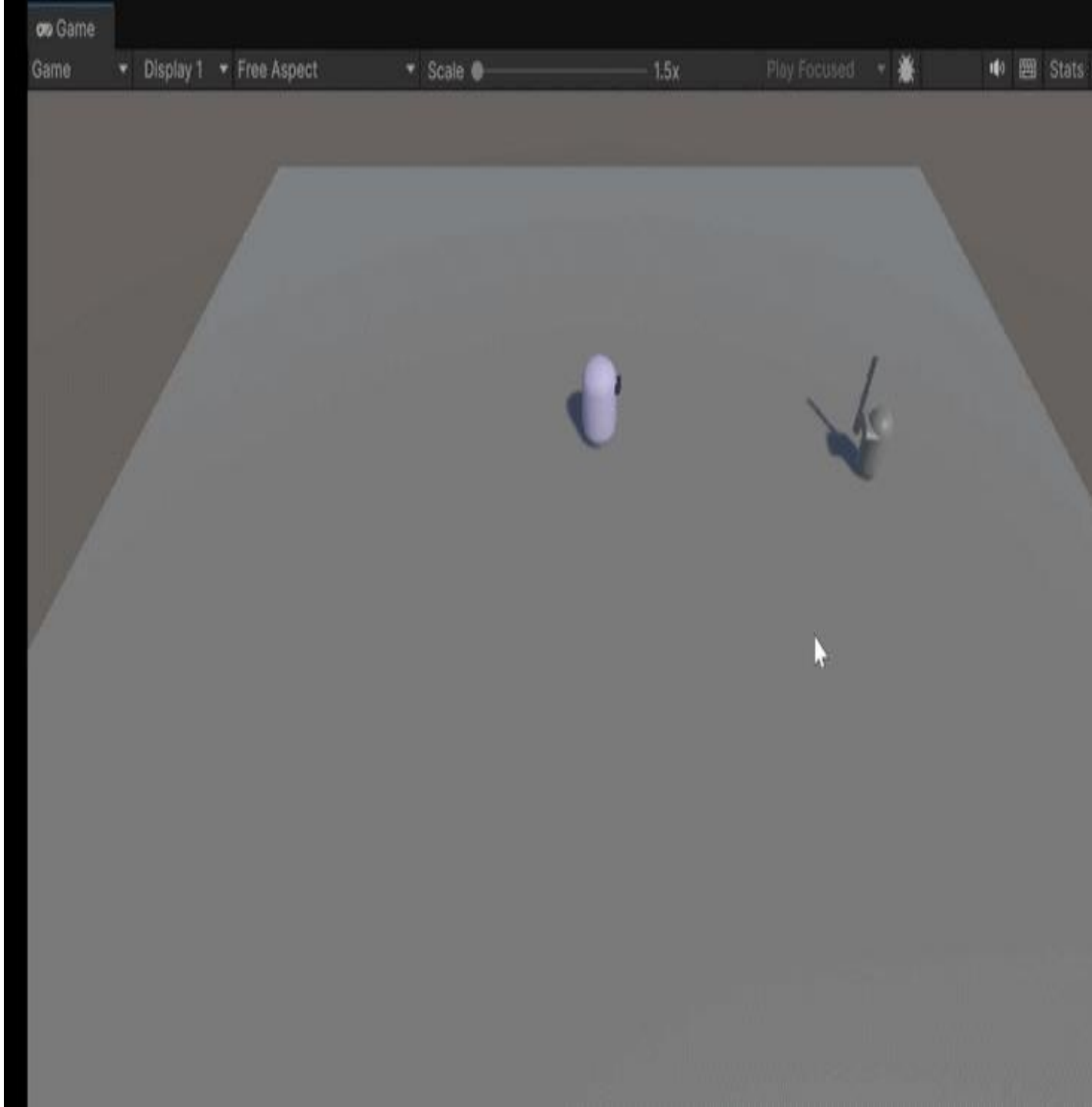
Create a boss in a game that learns how to fight the player

- Need within the Soulslike genre
  - Players are seeking harder and harder challenges
  - Developers need interesting and adaptive enemy behavior
- Trainable machine learning model dictates boss decisions
  - Allows for more complex fight decisions and fights
  - Circumvents manual coding of boss AI



Overview

- Created a fighting game in the Unity Engine.
  - Player with a limited move set.
  - Boss with wide range of attacks.
- Compiled game state as a vector.
- Trained boss bots by letting them fight simulated player bots.
- The models were created with the ML-Agents Package of Unity, which uses Proximal Policy Optimization (PPO) neural network (ensures stable and smooth learning).



Challenge 1: Overfitting

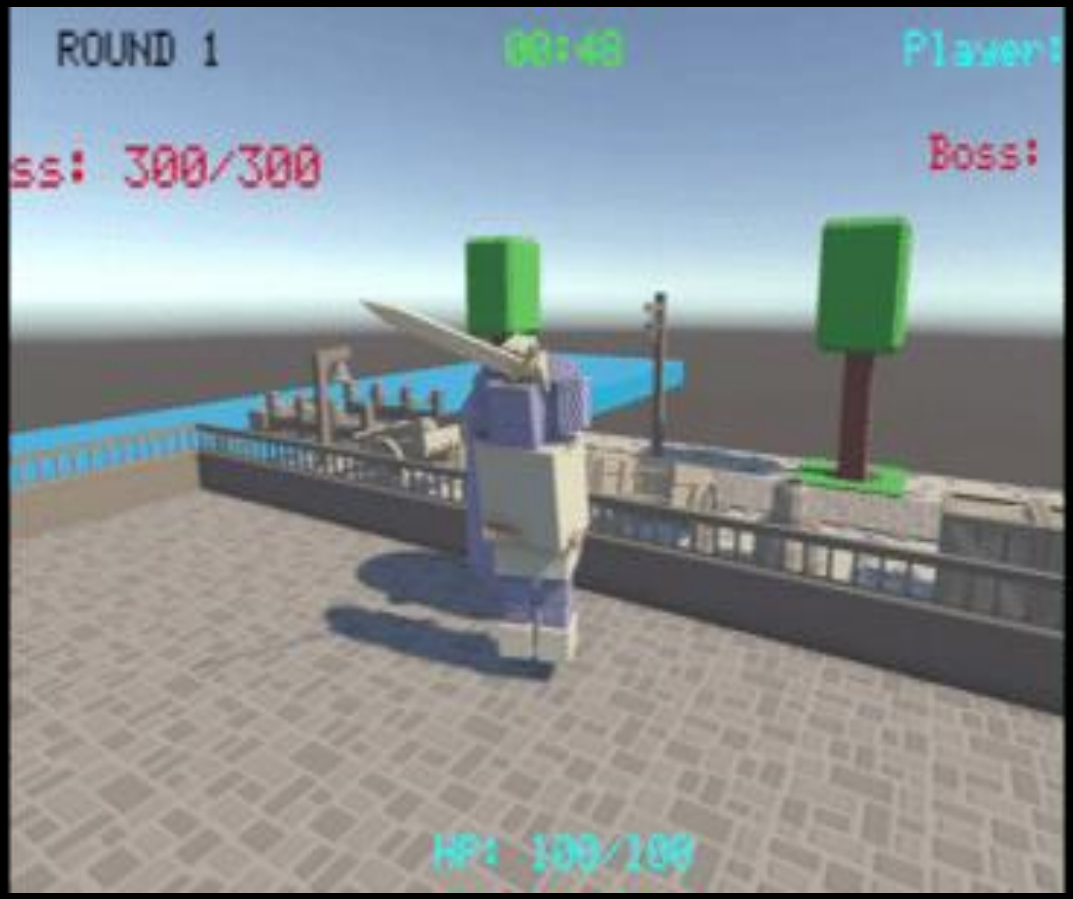
Problem

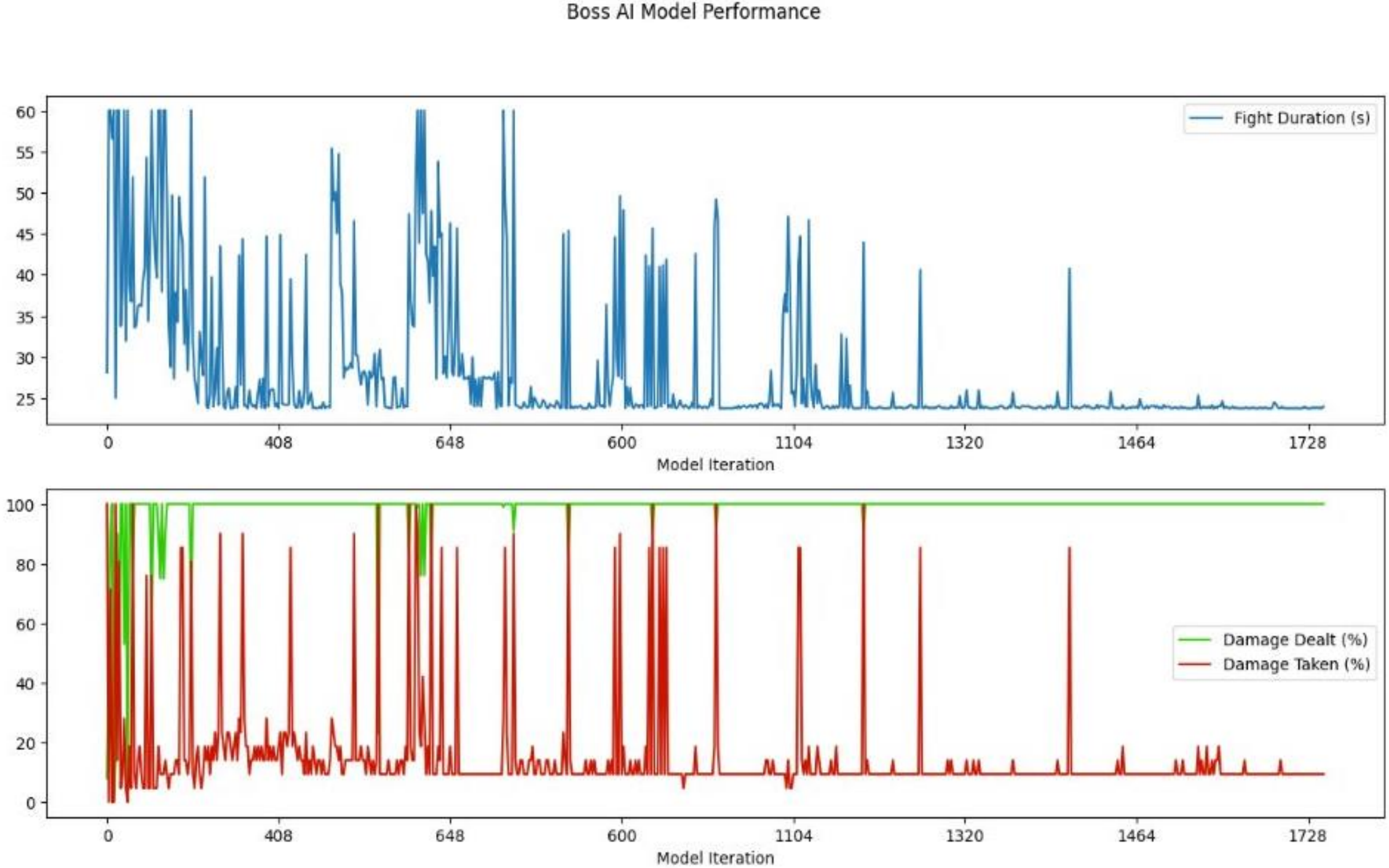
The boss bot easily beat the first hard-coded player bot, but it failed miserably against human players.

Solution

More Player Bots

- More Hardcoded Types: Simple, Coward, Aggressive.
- Imitation Learning: Train with human fighting samples.

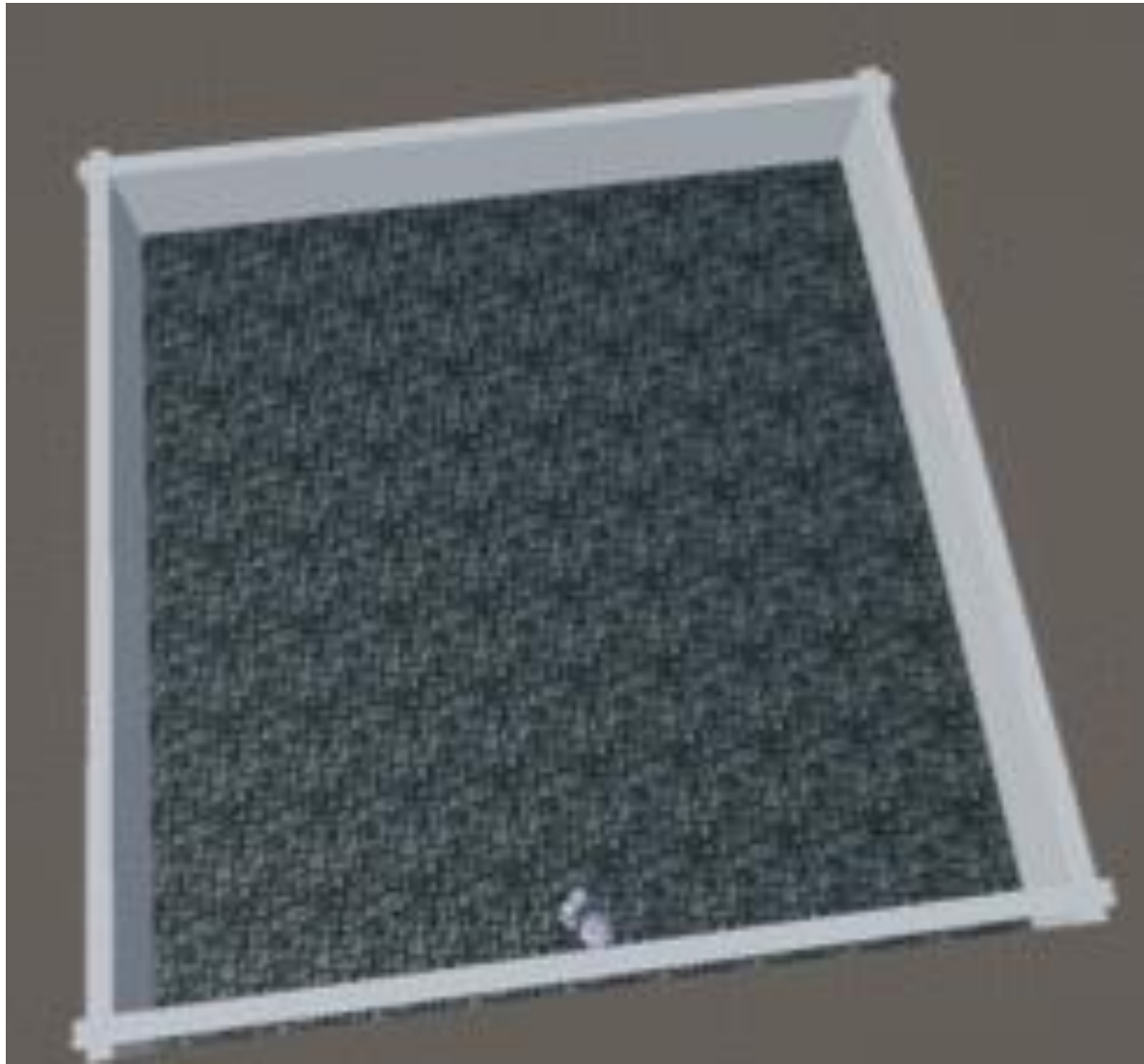



Boss AI Model Performance
Fight Duration (s)
60
55
50
45
40
35
30
25
0
408
648
600
1104
1320
1464
1728
Model Iteration
Damage Dealt (%)
Damage Taken (%)
100
80
60
40
20
0
0
408
648
600
1104
1320
1464
1728
Model Iteration

Challenge 2: Reward Hacking

Problem

- The boss was punished when games end due to time-out, but less so than a defeat.
- The boss found a local maximum reward by playing very cowardly and reaching a draw to avoid losing.
- However, the boss would win if it played more aggressively.

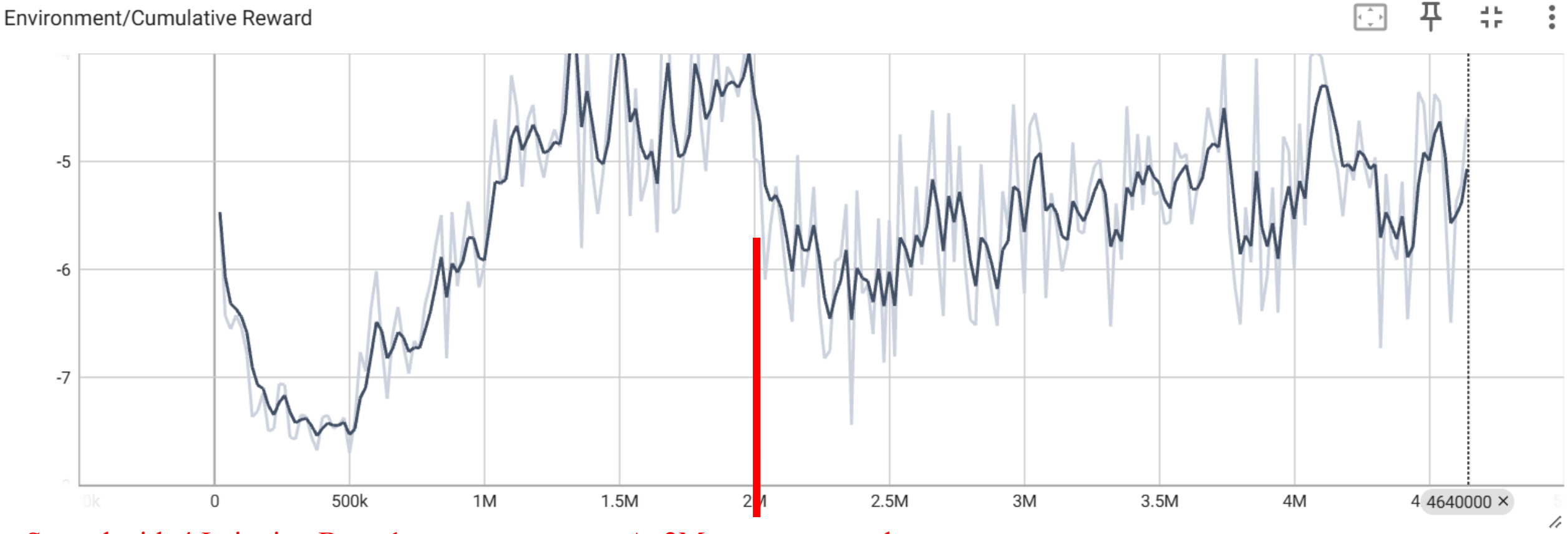


Solution

- Fine-tuned the reward function.
  - Make a tie-game equally as costly as losing.
  - Reward dealing damage higher than taking damage.

Trained with the following hyperparameters and rewards:

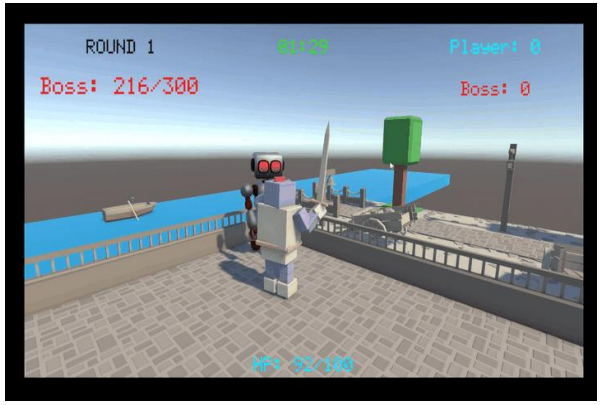
```
environment_parameters:
  win_reward: 5.0
  win_time_decay: 0.0
  loss_penalty: 5.0
  draw_penalty: 5.0
  damage_dealt_reward: 0.8
  damage_taken_penalty: 0.25
  time_penalty_per_second: 0.05
  reaction_delay: 0.01
```



Started with 4 Imitation Bots, 1 Aggressive Bot, 1 Cowardly Bot, 2 Default Bot

At 2M steps, swapped to 4 Aggressive Bots, 4 Imitation Bots

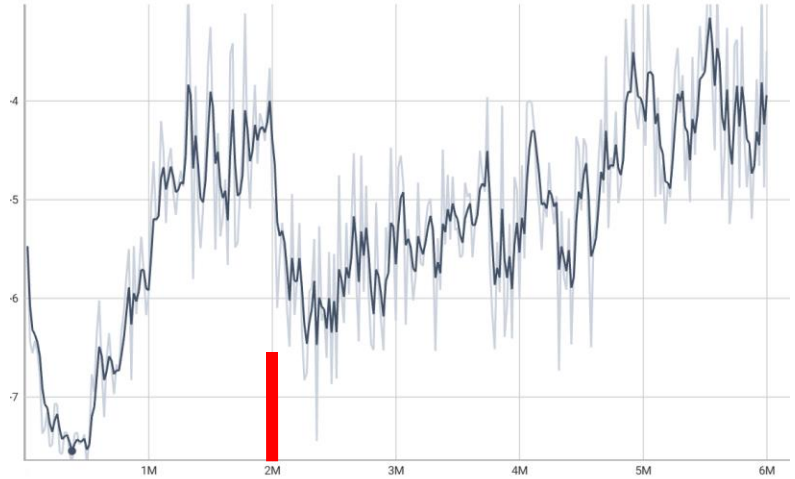
Final Boss Model



The current best-performing boss model was named andreBoss3, since it was the third iteration of a model trained by Andre.

The model used:

- 2 Layers
- 256 nodes per layer
- The previously mentioned reward changes



Graph of cumulative reward over time

More complex models tended to act strangely, with actions like repeatedly using the projectile attack.

By fighting the model ourselves, we found it to be the most challenging and engaging, with moderate difficulty and focus needed to beat it.

```
environment_parameters:
  win_reward: 5.0
  win_time_decay: 0.0
  loss_penalty: 5.0
  draw_penalty: 5.0
  damage_dealt_reward: 0.8
  damage_taken_penalty: 0.25
  time_penalty_per_second: 0.05
  reaction_delay: 0.01
```


ROUND 2
Player: 1
Boss: 12/300
Boss: 0
HP: 55/100


Thank You